in Table 2.

An example with EditText and TextView is illustrated in the following code snippet:

```
<LinearLayout android:orientation="horizontal"
        ... >
    <EditText android:id="@+id/age"
        android:nextFocusDown="@+id/town"
        ... />
    <TextView android:id="@+id/town"
        android:focusable="true"
        android:text="Pondicherry"
        android:nextFocusUp="@id/age"
        ... />
</LinearLayout>
```

Table 2: Android focus management attributes

Attributes	Description
android:nextFocusDown	Refers to the next element which will receive focus when the user navigates down
android:nextFocusLeft	Refers to the next element which will receive focus when the user navigates to the left
android:nextFocusRight	Refers to the next element which will receive focus when the user navigates to the right
android:nextFocusUp	Refers to the next element which will receive focus when the user navigates up

Step 3: Implementation of accessibility interfaces needs to be done in case of custom controls. The corresponding support libraries should be utilised to implement these features.

Step 4: For all the audio-only notifications/alerts, a visual alternative should be provided so that persons with hearing disabilities can use the application efficiently. The opposite scenario also needs to be handled to assist persons with visual impairments.

Step 5: Testing of accessibility features such as *Talkback* or *Explore by Touch* needs to be carried out with these features activated.

The *TalkBack* service should be activated using the following steps:

- Go to *Settings -> Accessibility category*
- Select *Accessibility* and *Talkback* to enable

Explore by Touch should be utilised in devices with Android version 4.0 and above. The *Explore by Touch* mode enables the users to drag their finger around the user interface to hear audible representations. Further instructions on *Explore by Touch* are available at *https://support.google.com/accessibility/android/answer/6006598*.

Explore by Touch must be enabled with the following steps:

- Go to *Settings -> Accessibility category*
- Enable *TalkBack*
- Return to *Accessibility* and enable *Explore by Touch*

The focus management explained in the previous section should be tested with Android Emulator using a simulation of the directional controller.

Google Accessibility Scanner

Google has recently launched a tool to make the task of accessibility testing of an app simple. This app is called the Accessibility Scanner and it can be downloaded from the Google Play Store *https://play.google.com/store/apps/details?id=com.google.android.apps.accessibility.auditor&hl=en*.

Accessibility Scanner scans Android applications and provides suggestions to improve them. Sample tasks performed by the Accessibility Scanner are listed below:

- Identifying small touch targets and providing instructions to enlarge them.
- Increasing the contrast of the user interface elements.
- Providing content descriptions.

The steps involved in using the Accessibility Scanner are listed below:

- Go to *Settings -> Accessibility*
- Enable *Accessibility Scanner*
- Go to the app that you want to scan for accessibility
- Tap the *Accessibility Scanner* button to start the scan of the interface

Apart from the legal requirement of making applications accessible, it is definitely an ethical practice for any developer to build applications that can be used by everyone, irrespective of their physical disability. The growing importance of accessibility can be gauged by facts such as the moving of the accessibility vision settings to the welcome screen in the upcoming Android N version *(https://googleblog.blogspot.in/2016/04/building-more-accessible-technology.html)*.

In the final analysis, building accessible applications is a win-win scenario for both developers and end users. END

References

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By: Dr K.S. Kuppusamy

The author is assistant professor of computer science at the Pondicherry Central University. He has vast experience in teaching and research in academia and industry. He is interested in accessible computing, Web information retrieval, mobile computing, etc. He can be reached via mail at *kskuppu@gmail.com*.